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(54) EXTENDABLE SUB-FLOOR STORAGE COVER

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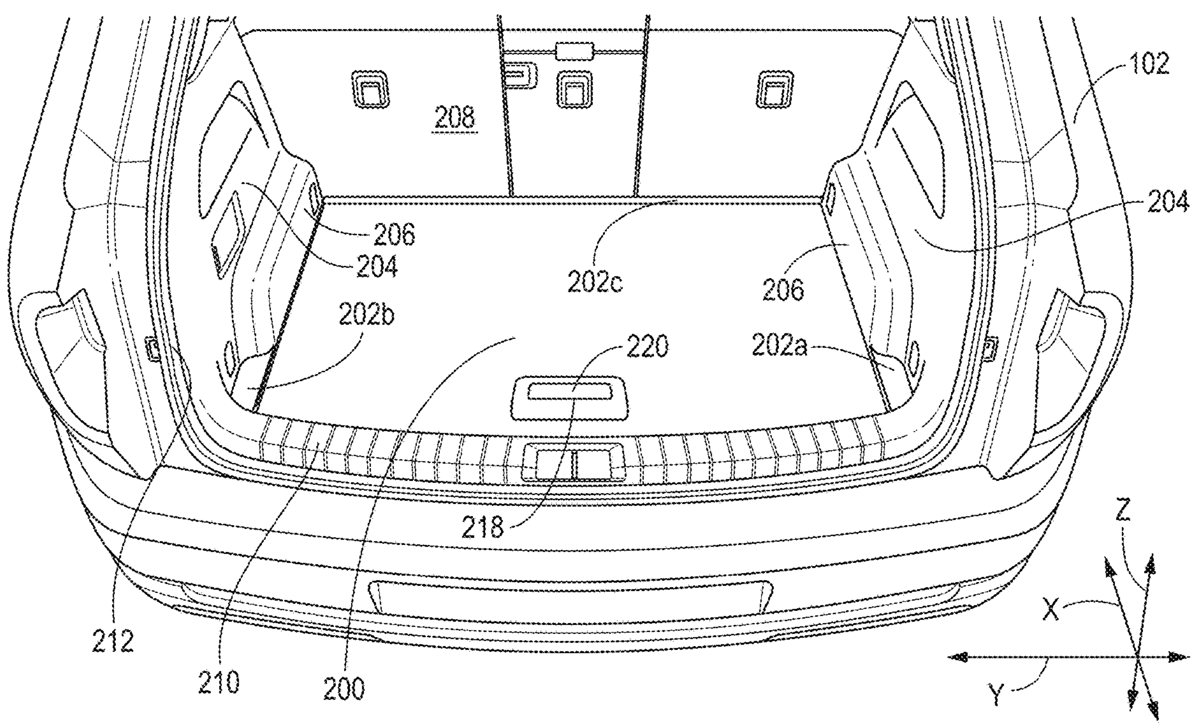
(57) **ABSTRACT**

A cover is configured to cover the sub-floor storage area of a vehicle and defines at least part of a load floor over the sub-floor storage area. Guides are secured to the vehicle body and define paths configured to guide movement of the cover between: a closed position in which the cover extends over the sub-floor storage area; an extended position in which the cover remains engaged with the guides and extends outward from the vehicle body; and an open position in which the cover is pivoted upwardly relative to the closed position. The guides may be grooves formed in the vehicle body and followers secure to the cover and are configured to engage the guides, such as rollers or slides. The guides may each include a first portion defining a path to the extended position, a second portion in which the followers seat when in the closed position.



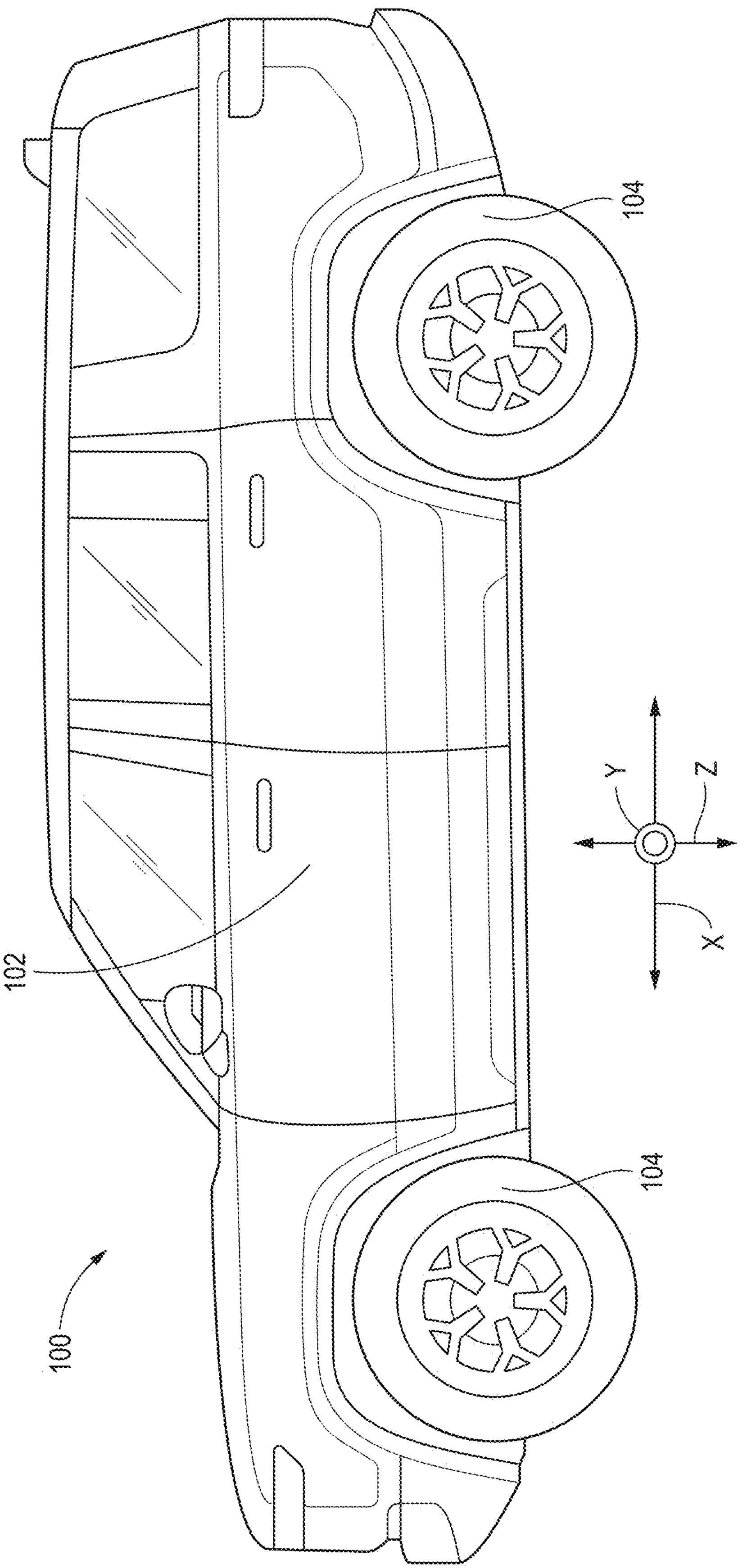


FIG. 1A

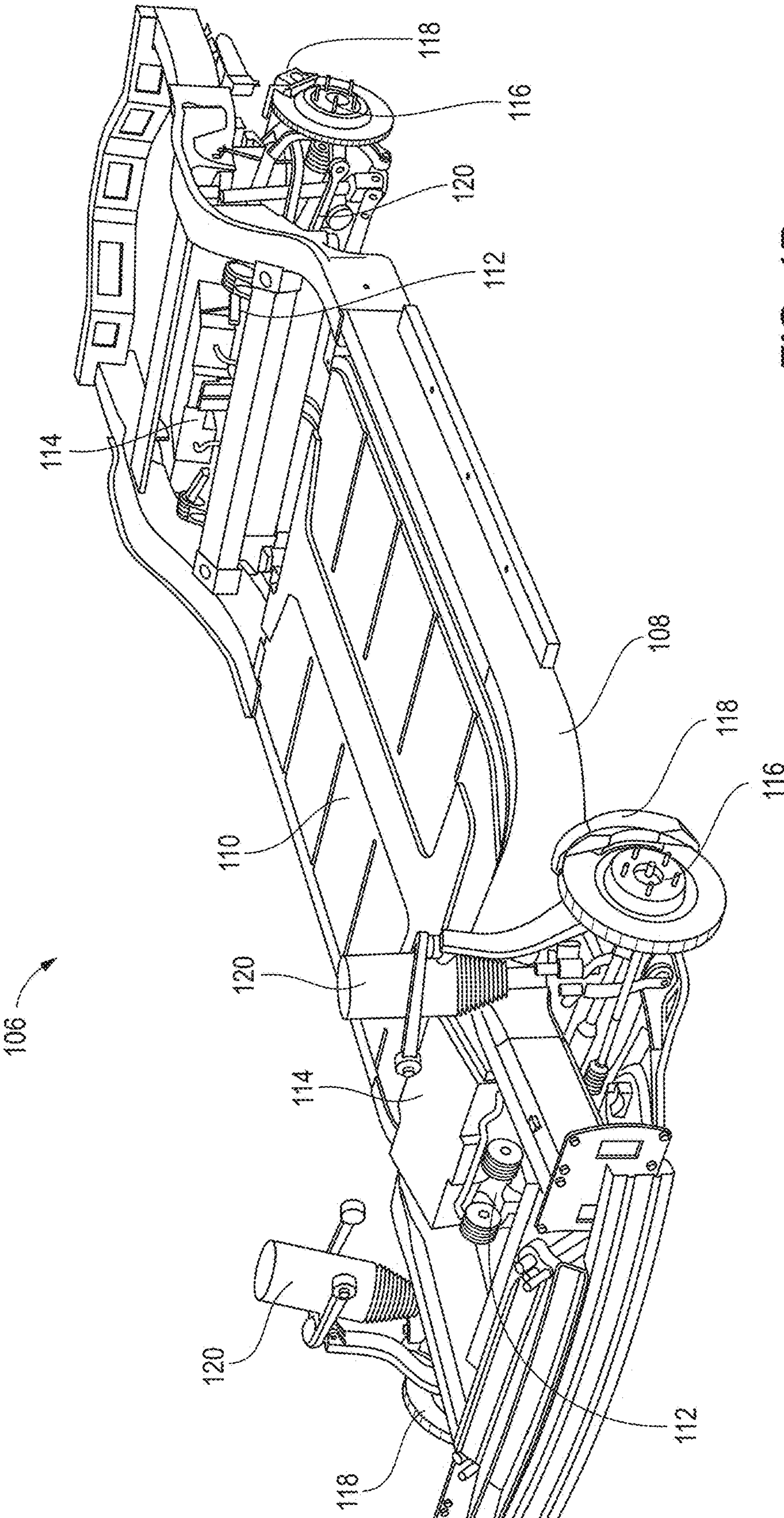


FIG. 1B

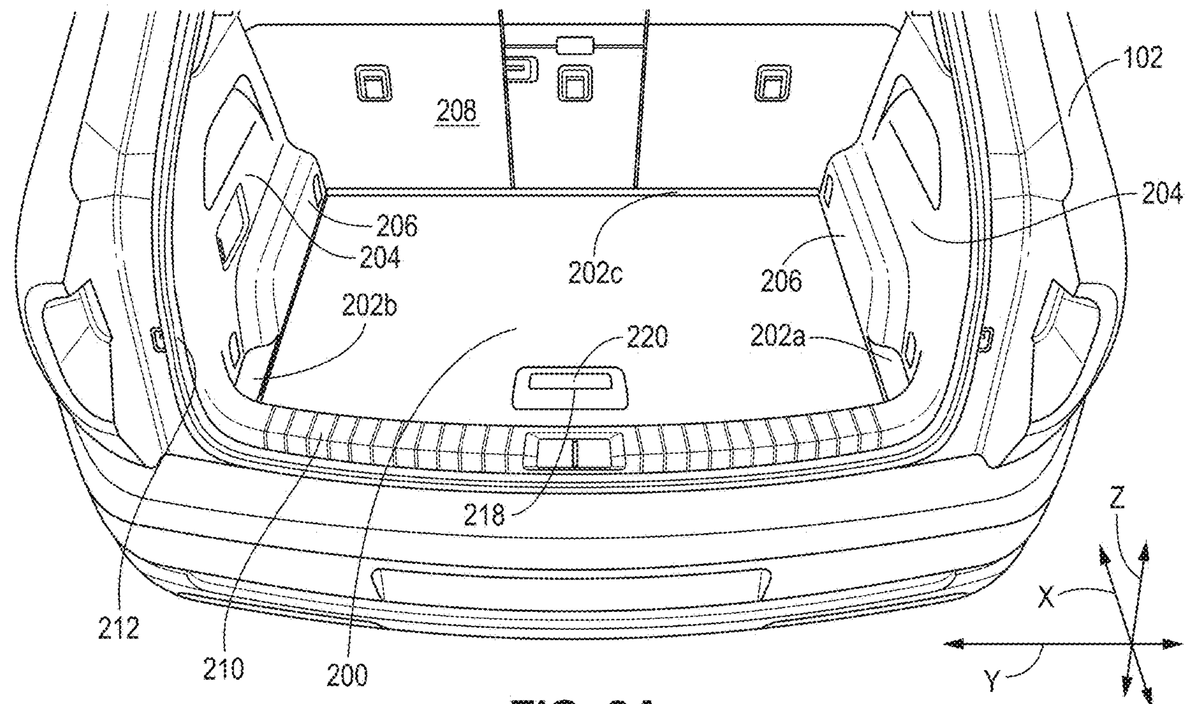

102
208
206
204
204
202c
206
202b
220
202a
218
212
210
200
Z
X
Y

FIG. 2A

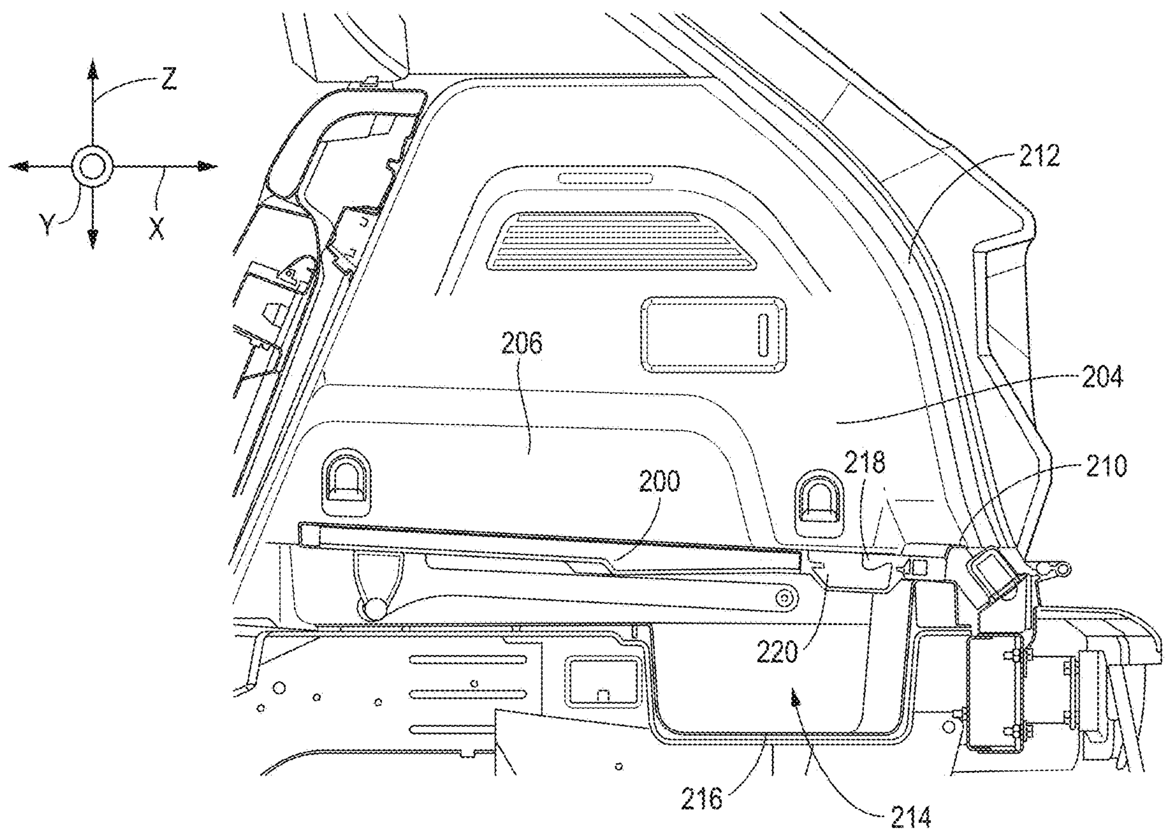

Z
Y
X
212
206
204
218
210
200
220
216
214

FIG. 2B

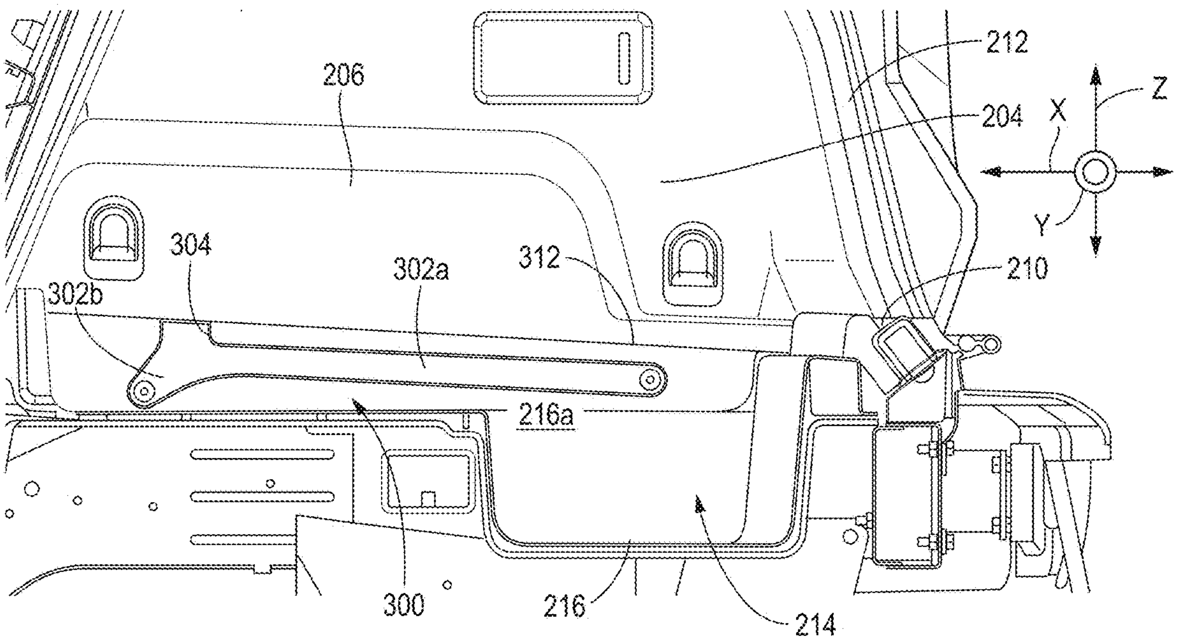

206
212
204
X
Z
Y
304
302a
312
302b
210
216a
300
216
214

FIG. 3

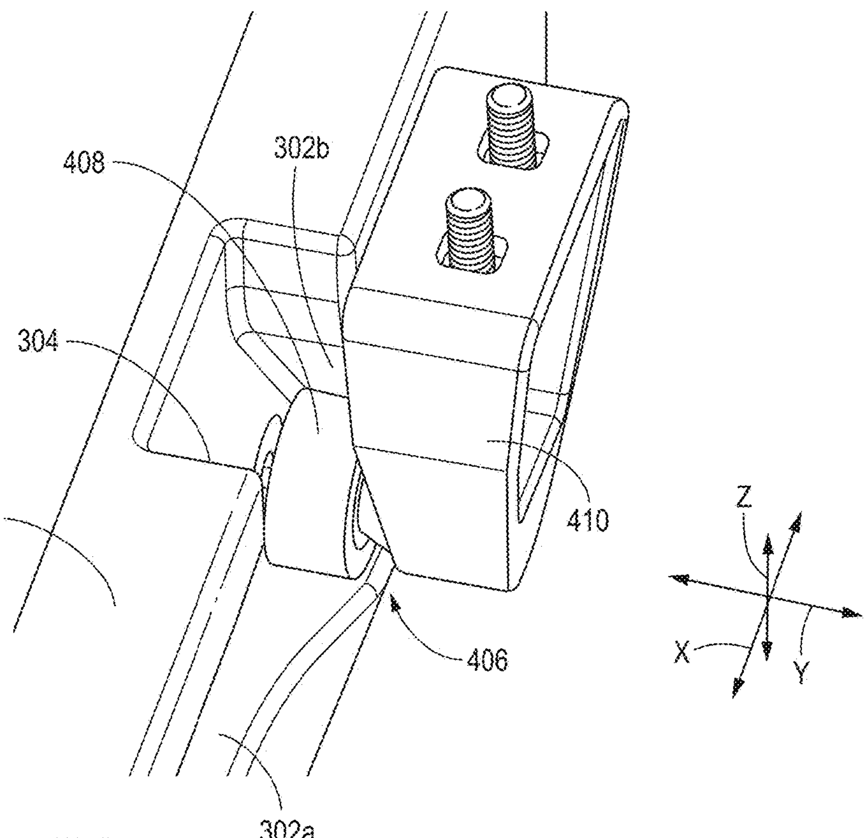

408
302b
304
312
410
Z
X
Y
406
302a

FIG. 4

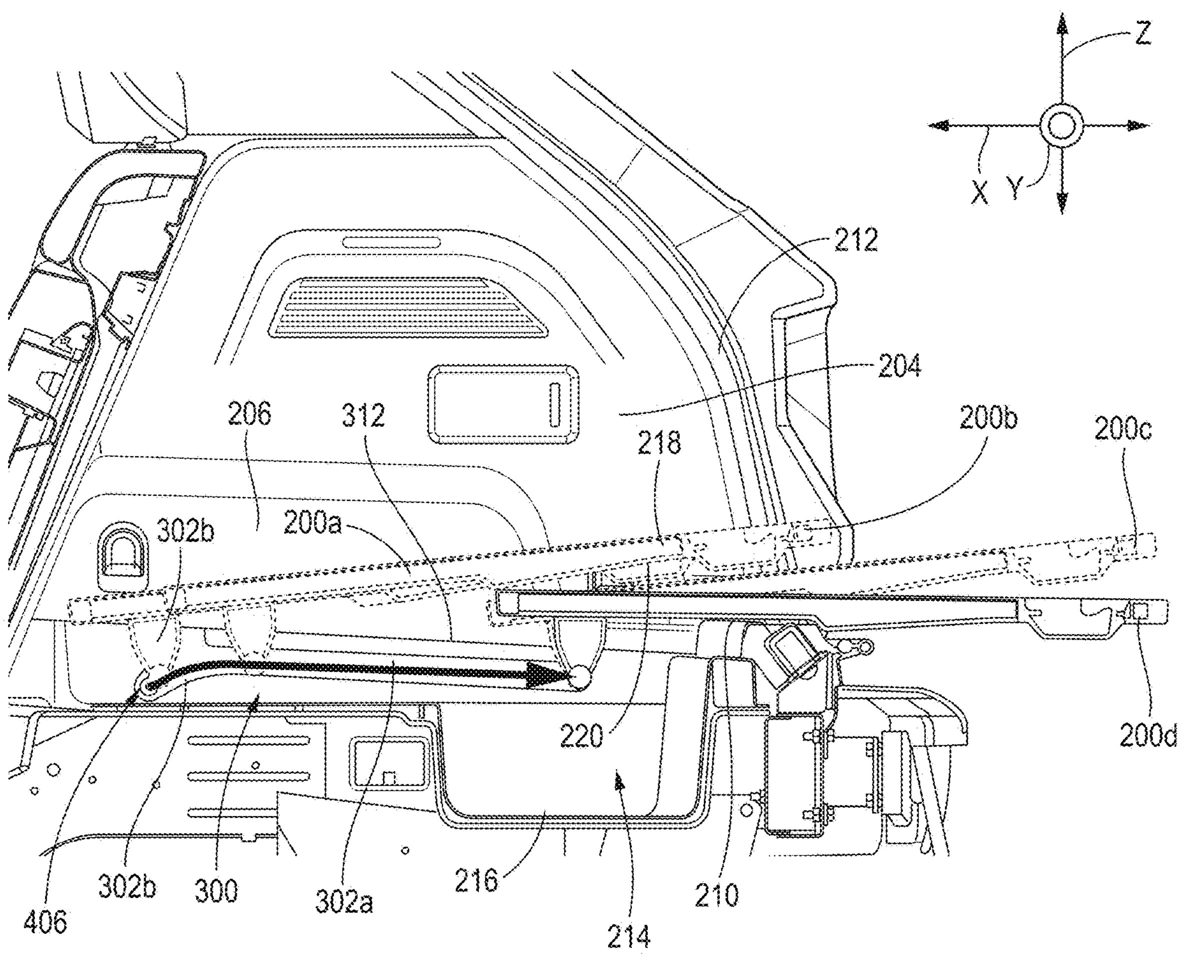

Z
X
Y
212
204
206
312
200b
200c
218
302b
200a
220
200d
406
302b
300
302a
216
210
214

FIG. 5

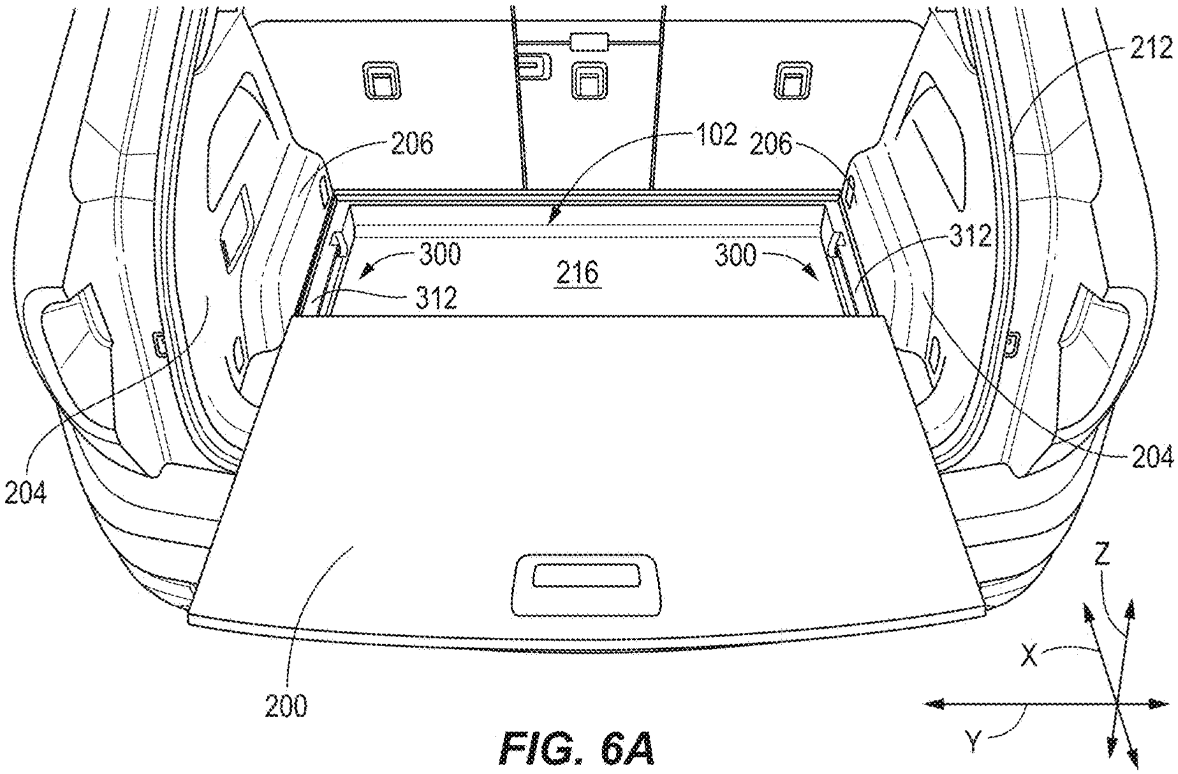

212
206
102
206
300
216
300
312
312
204
204
200
Z
X
Y

FIG. 6A

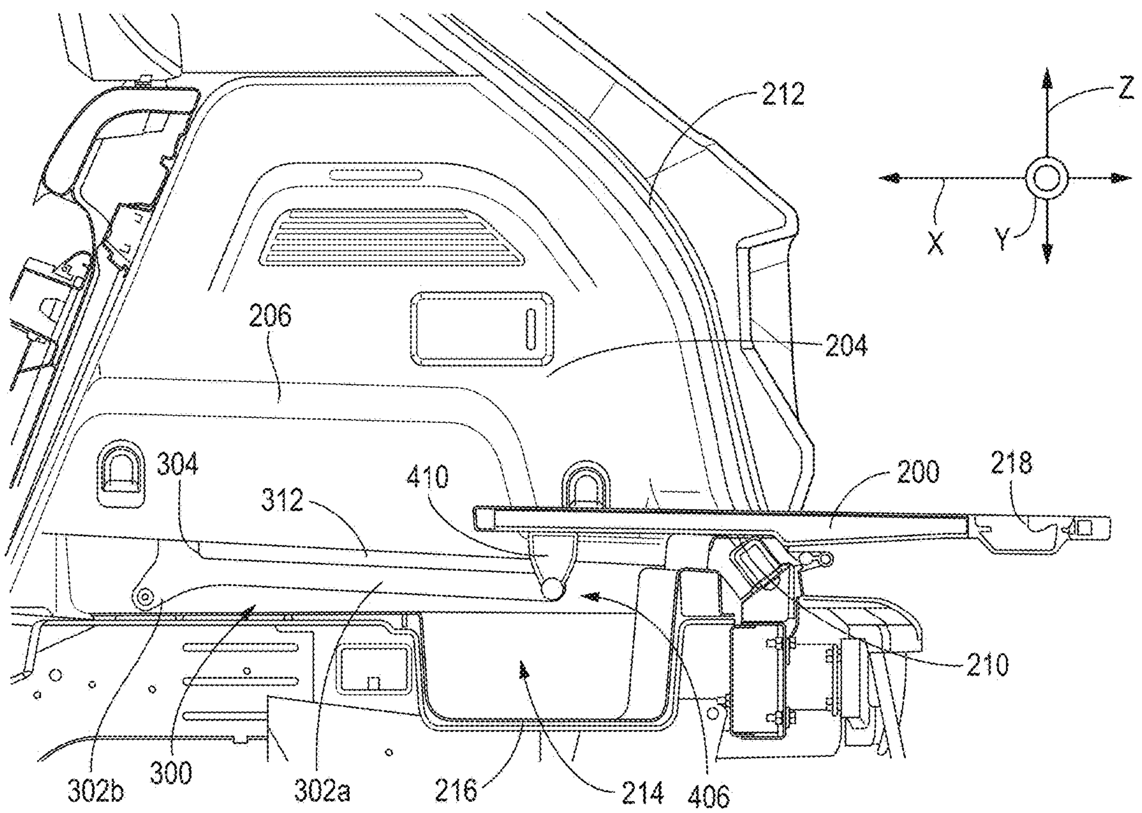
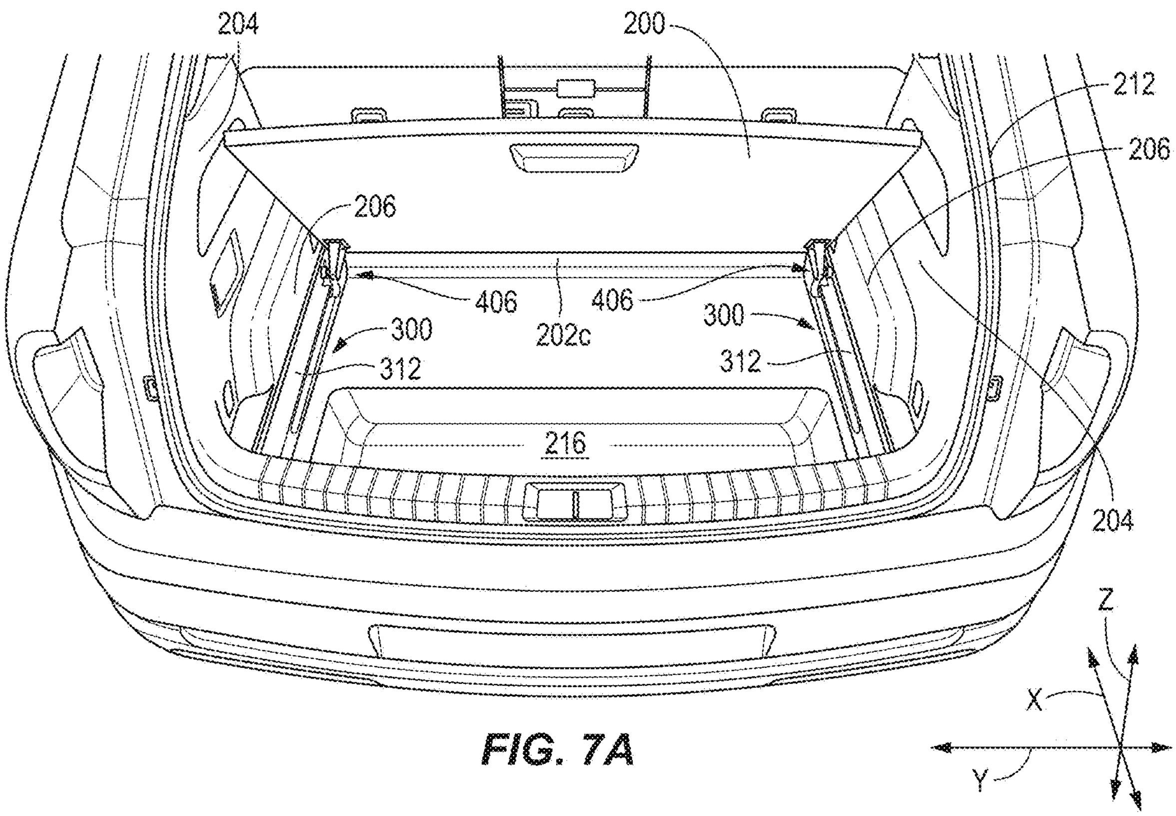
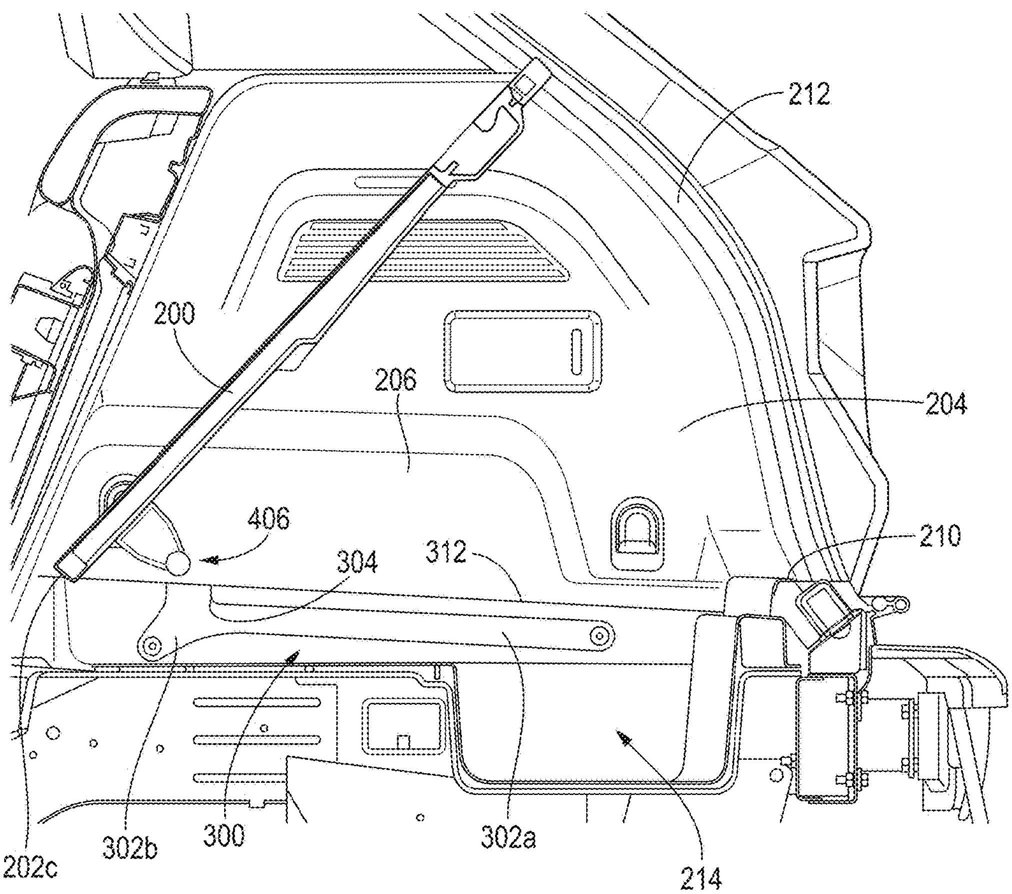

212
Z
X
Y
206
204
304
312
410
200
218
210
302b
300
302a
216
214
406

FIG. 6B



204
200
212
206
206
406
406
202c
300
300
312
312
216
204
Z
X
Y

FIG. 7A



212
200
206
204
406
304
312
210
202c
302b
300
302a
214

FIG. 7B

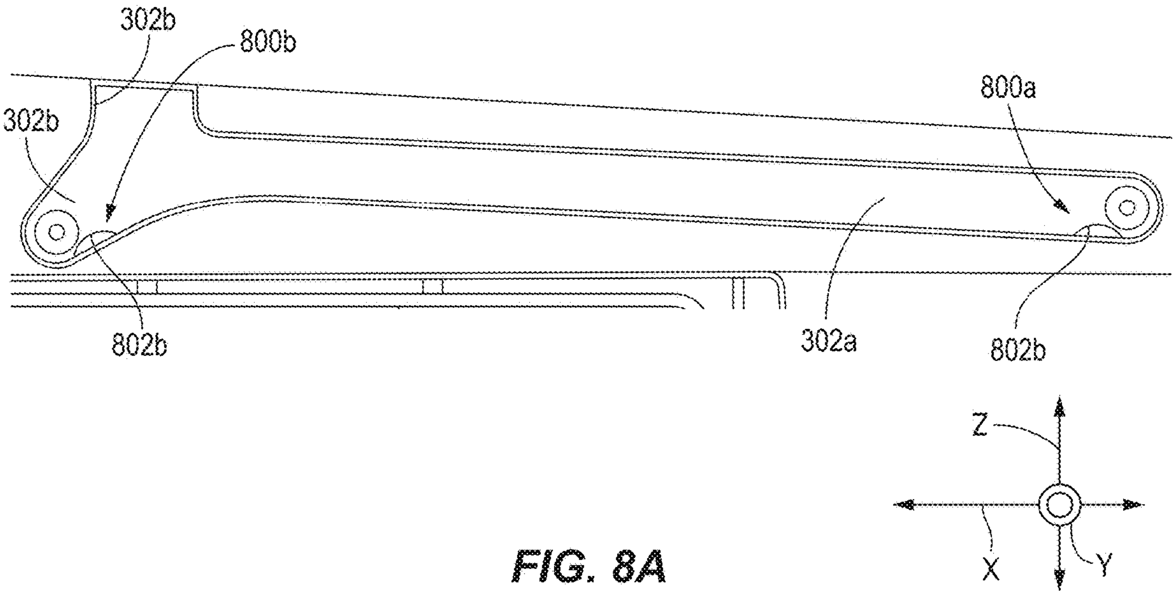

302b
800b
302b
800a
802b
302a
802b
Z
X
Y

FIG. 8A

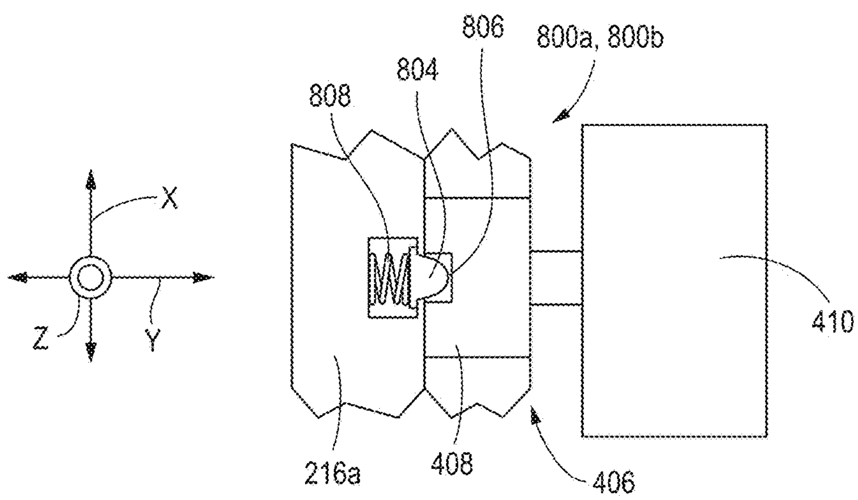

806
800a, 800b
804
808
X
410
Z
Y
216a
408
406

FIG. 8B

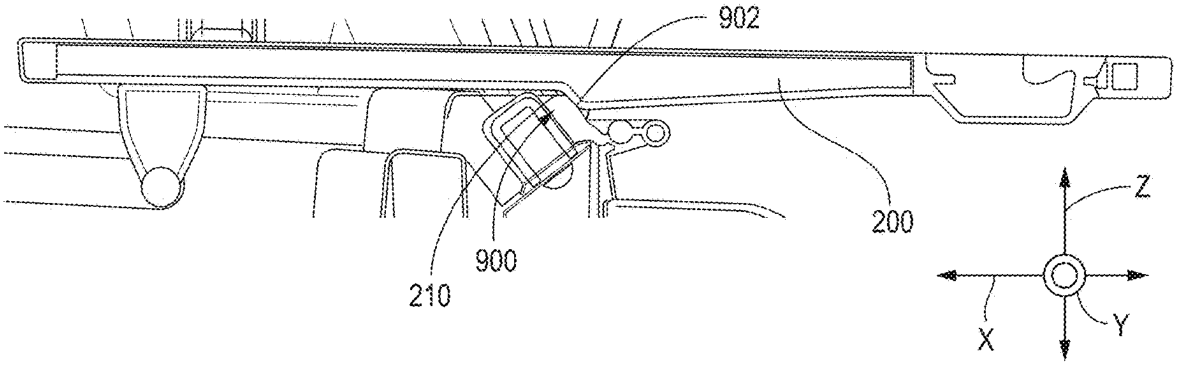


FIG. 9A

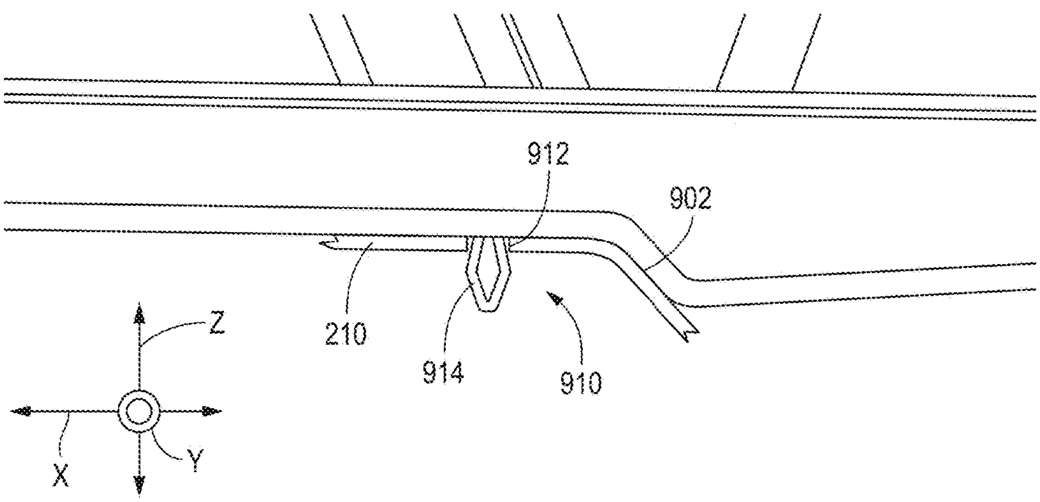


FIG. 9B

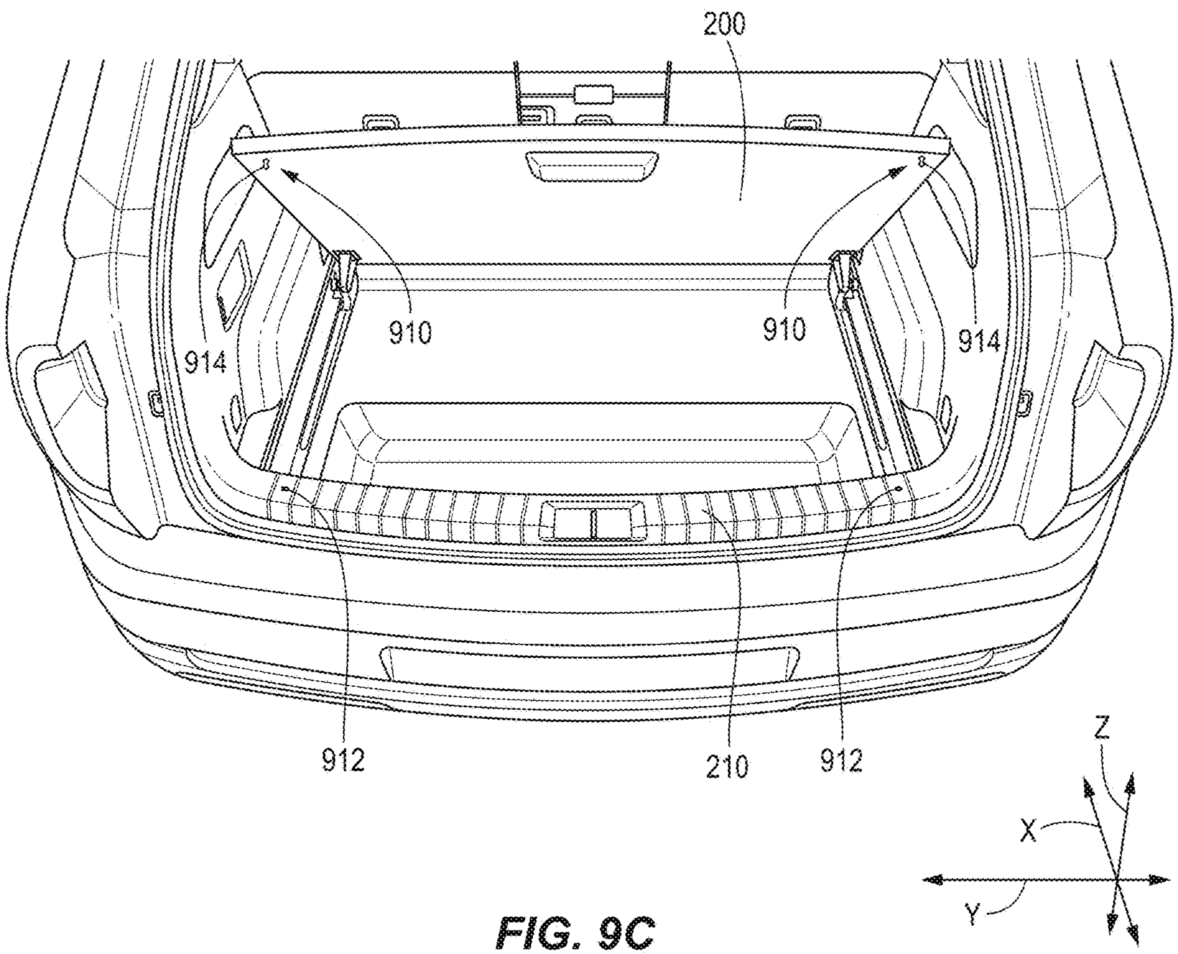


FIG. 9C

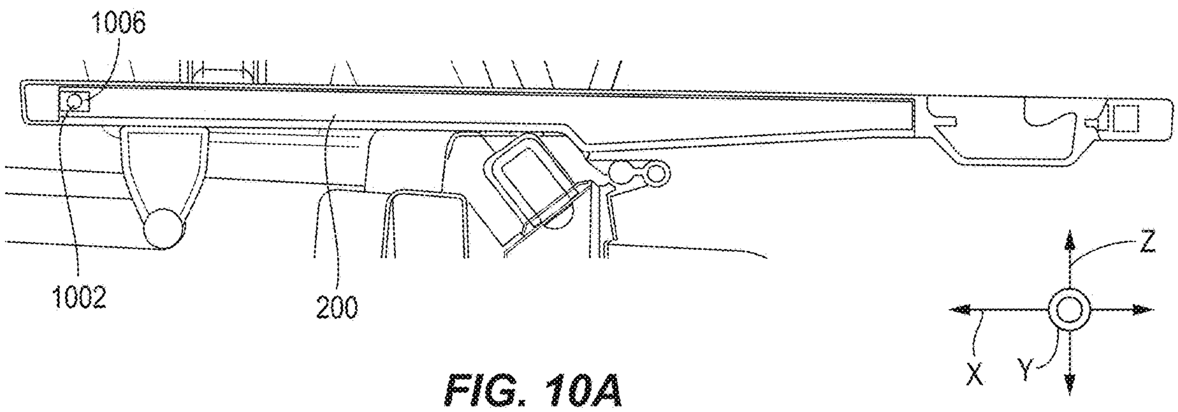


FIG. 10A

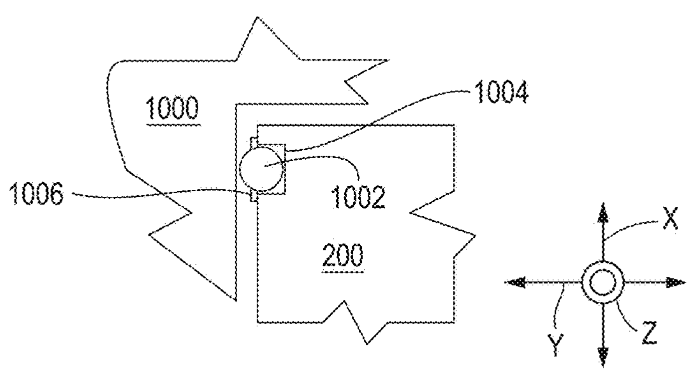


FIG. 10B

EXTENDABLE SUB-FLOOR STORAGE COVER

RELATED APPLICATION

[0001] This application claims the benefit of U.S. Application Ser. No. 63/559,086 filed Feb. 28, 2024, and entitled EXTENDABLE SUB-FLOOR STORAGE COVER.

INTRODUCTION

[0002] The present disclosure relates to a cover for a sub-floor storage volume of a vehicle.

SUMMARY

[0003] The present disclosure describes an approach for implementing a cover for sub-floor storage. In one aspect, a vehicle body defines a sub-floor storage area. A cover is configured to cover the sub-floor storage area and defines at least part of a load floor over the sub-floor storage area. Guides are secured to the vehicle body and define paths configured to guide movement of the cover between: a closed position in which the cover extends over the sub-floor storage area; an extended position in which the cover remains engaged with the guides and extends outward from the vehicle body; and an open position in which the cover is pivoted upwardly relative to the closed position.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1**A illustrates an example vehicle that may be used in accordance with certain embodiments.

[0005] FIG. **1**B illustrates a chassis of a vehicle having multiple drive units that may be used in accordance with certain embodiments.

[0006] FIG. **2**A is an isometric view of a sub-floor storage cover in a closed position in accordance with certain embodiments.

[0007] FIG. **2**B is a cross-sectional side view of the sub-floor storage cover in a closed position in accordance with certain embodiments.

[0008] FIG. **3** is a side view of a guide for facilitating opening and extending of the sub-floor storage cover in accordance with certain embodiments.

[0009] FIG. **4** is an isometric view of a roller within the guide in accordance with certain embodiments.

[0010] FIG. **5** is a cross-sectional side view illustrating a process for extending the sub-floor storage cover in accordance with certain embodiments.

[0011] FIG. **6**A is an isometric view illustrating the sub-floor storage cover in an extended position in accordance with certain embodiments.

[0012] FIG. **6**B is a cross-sectional side view of the sub-floor storage cover in the extended position in accordance with certain embodiments.

[0013] FIG. **7**A is an isometric view illustrating the sub-floor storage cover in an open position in accordance with certain embodiments.

[0014] FIG. **7**B is a cross-sectional side view of the sub-floor storage cover in the open position in accordance with certain embodiments.

[0015] FIG. **8**A is a side view illustrating the guide with integrated retention features in accordance with certain embodiments.

[0016] FIG. **8**B is a top view illustrating an alternative retention feature in accordance with certain embodiments.

[0017] FIG. **9**A is a cross-sectional side view illustrating nesting of the sub-floor storage cover on a scuff plate in accordance with certain embodiments.

[0018] FIG. **9**B is a cross-sectional side view illustrating a retention feature for retaining the sub-floor storage cover relative to the scuff plate in accordance with certain embodiments.

[0019] FIG. **9**C is an isometric view illustrating placement of retention features according to FIG. **9**B in accordance with certain embodiments.

[0020] FIG. **10**A is a side view illustrating a roller for facilitating sliding in accordance with certain embodiments.

[0021] FIG. **10**B is a top cross-sectional view showing the roller of FIG. **10**A.

DETAILED DESCRIPTION

[0022] A vehicle defines sub-floor storage. A cover is positionable over the sub-floor storage and forms at least part of a load floor of the vehicle. Guides are secured to the vehicle, such as to sidewalls of the sub-floor storage. The guides engage followers secured to the cover and guide the cover between closed, extended, and open positions, such as: a closed position in which the cover extends over the sub-floor storage area; an extended position in which the cover remains engaged with the guides and extends outward from the vehicle body; and an open position in which the cover is pivoted upwardly relative to the closed position.

[0023] FIG. **1**A illustrates an example vehicle **100** in which the approach described herein may be implemented. As seen in FIG. **1**A, the vehicle **100** has a body **102** defining a passenger cabin and internal cargo area of the vehicle **100**. The body **102** may be configured as a sport utility vehicle (SUV), pickup truck including external bed, or other type of vehicle. The vehicle **100** further includes a plurality of road wheels **104**, such as four, that are driven to propel the vehicle **100** over a surface.

[0024] The vehicle **100** may be understood with respect to X, Y, and Z directions. The Z direction may correspond to the direction of gravity when the vehicle **100** is on a flat, level surface perpendicular to the direction of gravity. The X direction may correspond to the direction of the vehicle **100** when traveling in a straight line. The Y direction may be defined as perpendicular to the X and Z directions.

[0025] Referring to FIG. **1**B, the vehicle **100** may include a chassis **106** including a frame **108** providing a primary structural member of the vehicle **100**. The frame **108** may be formed of one or more beams or other structural members or may be integrated with the body of the vehicle (e.g., unibody construction).

[0026] In embodiments where the vehicle **100** is a battery electric vehicle (BEV) or possibly a hybrid vehicle, a large battery **110** is mounted to the chassis **106** and may occupy a substantial (e.g., at least 80 percent) of an area within the frame **108**. For example, the battery **110** may store from 100 to 200 kilowatt hours (kWh). The battery **110** may be a lithium-ion battery or other type of rechargeable battery. The battery may be substantially planar in shape.

[0027] Power from the battery **110** may be supplied to one or more drive units **112**. Each drive unit **112** may be formed of an electric motor and possibly a gear train providing a gear reduction. In some embodiments, there is a single drive unit **112** driving either the front wheels or the rear wheels of the vehicle **100**. In another embodiment, there are two drive units **112**, each driving either the front wheels or the rear wheels of the vehicle **100**. In yet another embodiment, there are four drive units **112**, each drive unit **112** driving one of four wheels of the vehicle **100**.

[0028] Power from the battery **110** may be supplied to the drive units **112** by one or more sets of power electronics **114**, such as power electronics for each drive unit **112** or pair of drive units **112**. The power electronics **114** may include inverters configured to convert direct current (DC) from the battery **110** into alternating current (AC) supplied to the motors of the drive units **112**. The power electronics **114** further facilitate operation of the motors of the drive units as generators to provide regenerative braking. The power electronics **114** further facilitate the transfer of regenerative current to the battery **110**.

[0029] The drive units **112** are coupled to two or more hubs **116** to which wheels **104** may mount. Each hub **116** includes a corresponding brake **118**, such as the illustrated disc brakes. Each hub **116** is further coupled to the frame **108** by a suspension **120**. The suspension **120** may include metal or pneumatic springs for absorbing impacts. The suspension **120** may be implemented as a pneumatic or hydraulic suspension capable of adjusting a ride height of the chassis **106** relative to a support surface. The suspension **120** may include a damper with the properties of the damper being either fixed or adjustable electronically.

[0030] In the embodiment of FIG. **1**B, the vehicle **100** is a battery electric vehicle. However, a hybrid-electric vehicle or internal combustion engine vehicle may also benefit from the approach described herein. In particular, the vehicle **100** may be any vehicle having an internal or external storage area.

[0031] FIGS. **2**A and **2**B illustrate a sub-floor storage cover **200** (hereinafter "cover **200**") for use in a cargo area or other area of a vehicle. The cover **200** may define a substantially (e.g., within **2** centimeters of) planar surface, and the substantially planar surface may have a substantially rectangular shape (e.g., within 5 centimeters of a rectangular shape). The cover **200** may seat within a load floor of a vehicle. The load floor may be a generally (e.g., within 4 centimeters of) planar surface that is substantially (e.g., within 10 degrees of) parallel to the Y and X directions. The load floor may be the exposed bed of a pickup truck or an enclosed floor of a passenger or cargo vehicle. In some embodiments, such as that illustrated in FIGS. **2**A and **2**B, the cover **200** forms substantially all (e.g., at least 80, 90, or 95 percent) of the load floor. However, areas **202***a*, **202***b*, **202***c* around the cover **200** may also form part of the load floor. In the Y direction, the load floor may extend between one or both of internal side walls **204** and internal wheel well covers **206**. In the X direction, the load floor may extend between seats **208** and lip **210**, such as a scuff plate or other rearwardly positioned portion of the vehicle body **102**. The load surface may be positioned below the lip **210** along the Z direction, such as between 5 and 25 centimeters below the lip **210**.

[0032] The load floor may be accessible through an opening **212** or be exposed as in the case of a pickup truck bed. In some embodiments, the lip **210** forms a lower edge of the opening **212**. The opening **212** may be covered by a liftgate hinged at the top of the opening **212**, a tailgate hinged at the bottom of the opening **212**, or door hinged at one or both sides of the opening **212**.

[0033] As shown in FIG. **2**B, the cover **200** covers a sub-floor storage area **214**. The sub-floor storage area **214** may be available for storing cargo and may additionally or alternatively store equipment such as a spare tire, tire repair kit, tools, vehicle accessories, or the like. The sub-floor storage area **214** may have a volume of at least 20, 40, or 60 liters. In the illustrated embodiment, the sub-floor storage area **214** may be defined by the cover **200** and a sub-floor **216** formed by a lower body panel of the body **102** or other component of the vehicle **100**.

[0034] The cover **200** may include a handle **218** to facilitate moving the cover **200** between the closed position shown in FIGS. **2**A and **2**B and extended and open positions as discussed below. The handle **218** may be positioned within a recess **220** defined by the cover **200**. For example, the handle **218** and recess **220** may be formed in the cover **200** during formation of the cover **200** by molding or other process. The handle **218** may be positioned flush with or below the upper surface of the cover **200** when the cover is in the closed position.

[0035] The cover **200** may be made of a plastic such as acrylonitrile butadiene styrene (ABS), polypropylene, nylon, polyvinyl chloride (PVC), or other elastomer. The plastic may include glass fill, such as between 10 and 50 percent. The cover **200** may also be made of a composite material, such as fiberglass, carbon fiber, or KEVLAR composite material. The cover **200** may include stiffening elements, such as ribs formed thereon and/or therein (e.g., an internal honeycomb structure) to resist bending and failure of the cover **200** under load.

[0036] Referring to FIGS. **3** and **4**, movement of the cover **200** between the closed, extended, and open positions may be facilitated by a guide **300**. The guide **300** may be a groove secured by fasteners to a sidewall **216***a* extending upwardly from the sub-floor **216**. The guide may be secured to the sidewall **216***a* due to being formed therein, such as by molding. The guide **300** may alternatively be implemented as a rail secured to or formed on the sidewall **216***a*. The guide **300** may have multiple portions **302***a*, **302***b*. A first portion **302***a* guides the cover **200** between the closed position and the extended position. The second portion **302***b* guides the cover **200** between the closed position and an open position and between the closed position and the extended position. The guide **300** may define an inlet **304** at a junction between the first and second portions **302***a*, **302***b*.

[0037] The guide **300** may be engaged by a follower **406**. The follower **406** is substantially constrained to follow the first and second portions **302***a*, **302***b*. As used herein "substantially constrained" may be understood to mean constrained to remain within 3, 2, or 1 centimeters from a path defined by the first and second portions **302***a*, **302***b*. The follower **406** may be embodied as the illustrated roller **408** inserting into a guide **300** embodied as a groove. In other embodiments, the follower **406** may be implemented as a protrusion made of or coated with a smooth polymer such as nylon, TEFLON, ultra-high molecular weight (UHMW), or other low-friction material. The follower **406** may be a smooth protrusion inserting into the groove. Where the guide **300** is a rail, the follower **406** may be a pair of wheels on either side of the rail (e.g., along the Z direction), a pair of protrusions positioned on either side of the rail, or other type of follower. The follower **406** may be secured to the cover **200** by a bracket **410**. The bracket **410** may be secured to the cover **200** by fasteners or by co-molding with the cover **200**. In some embodiments, the follower, bracket **410**, and cover **200** are formed monolithically, such as by co-molding.

[0038] The portion **302***a* may substantially constrain the follower **406** to traverse a straight path substantially (e.g., within 15 degrees of) parallel to the X direction. As is apparent in FIG. **3**, the portion **302***a* may alternatively slope downwardly with proximity to the opening **212** along the X direction. For example, the portion **302***a* may slope downwardly at an angle of between 5 and 20 degrees. The angle of the portion **302***a* may be substantially (e.g., within 1 degree of) identical to an angle of the load floor in the plane defined by the X and Z directions.

[0039] The portion **302***b* may substantially constrain the follower **406** to traverse a path at a substantial angle relative to the X direction in a plane defined by the X and Z directions, such as an angle of between 30 and 60 degrees, such as between 40 and 50 degrees, such as between 43 and 47 degrees. The path defined by the portion **302***b* may slope downwardly with distance from the inlet **304**, e.g., with distance away from the opening **212**.

[0040] The inlet **304** may be positioned at a junction between the portion **302***a*, **302***b* and may be wider than the portion **302***a*, e.g., wider in the X direction than a width of the portion **302***a* in the Z direction. The widened inlet **304** may therefore facilitate insertion and removal of the follower **406**.

[0041] In the illustrated embodiment, the portion **302***a* is much longer than the portion **302***b*. For example, the portion **302***b* may be no longer, or only slightly (e.g., less than 2 centimeters) longer than required to provide clearance for the follower **406** when the cover **200** is in the closed position. In contrast, the portion **302***a* may have a length that is between 0.5 and 0.8 times the length of the cover **200** in the plane defined by the X and Z directions when the cover **200** is in the closed position. The length of the portion **302***a* may define the extent that the cover **200** may be extended when in the extended position as described below.

[0042] The portion **302***b* may have a tapered shape that narrows with distance from the inlet **304**. The tapered shape may facilitate a range of motion of the follower **406** in the plane defined by the Y and X directions when inserted into the inlet **304** and into the portion **302***b*. The tapered shape may further facilitate movement of the follower **406** when the cover **200** is pivoted into the open position as described in greater detail below.

[0043] FIGS. **3** and **4** further illustrate a shelf **312** that may support the cover **200** when in the closed position. The shelf **312** may be offset from and substantially (e.g., within 2 degrees of) parallel to the load floor, such as areas **202***a*, **202***b*, **202***c* of the load floor. The offset from the load floor may be substantially (e.g., within 5, 2, or 1 millimeters) equal to a thickness of the cover **200**, e.g., a portion of the cover **200** that rests on the shelf **312** when the cover **200** is in the closed position. The shelf **312** may be defined by the sidewall **216***a* of the sub-floor **216**, such as by molding into the sidewall **216***a*. The width of the shelf **312** in the Y direction may be selected to provide adequate support to the cover **200**. For example, for a maximum load (e.g., 200, 300, or 400 Newtons or more), the width of the shelf **312** may be selected such that the cover **200** will not slip off the shelf **312** due to bending of the cover **200**. In some embodiments, the inlet **304** extends to the shelf **312** and forms a gap between portions of the shelf **312** on either side of the inlet **304**.

[0044] The illustrated guide **300** and follower **406** of FIGS. **3** and **4** are shown for the right side of the storage area **214**. A corresponding guide **300** and follower **406** are also positioned on the left side of the storage area **214** in a mirrored configuration, e.g., mirrored about the plane defined by the Y and Z directions.

[0045] FIG. **5** illustrates an example process of moving the cover **200** from the closed position to the extended position. From the closed position, the user grasps the handle **218** and lifts upward, e.g., at least a component of exerted force upward along the Z direction. The cover **200** will move upward to position **200***a*. The user pulls the cover **200** to a height such that the rearward end of the cover **200** is positioned above the lip **210** as shown by position **200***b*. The user then pulls the cover **200** rearwardly, e.g., at least a component of force exerted rearward along the X direction. The follower **406** will then slide out of the portion **302***b*, past the inlet **304**, and into the portion **302***a*. The follower **406** will continue to follow the portion **302***a* until the follower **406** reaches the rearward end of the portion **302***a* as shown by position **200***c*. The user may then lower the cover **200** to rest on the lip **210**, as shown by position **200***d*, e.g., the extended position. In some embodiments, the cover **200** may be rested on the lip **210** at an intermediate position between the extended position and the closed position.

[0046] Transitioning the cover **200** back to the closed position may be performed by performing the reverse of the process of transitioning from the closed position to the extended position. The user lifts the cover **200** off the lip **210**, slides the cover **200** and follower **406** rearward until the follower **406** seats within the portion **302***b*, and then lowers the cover **200**, such as onto the shelf **312**. In some embodiments, one or more of the steps performed by a user as outlined above with reference to FIG. **5** may be performed by powered electronics, e.g., motors or other actuators. For example, the followers **406** may engage actuators that induce movement of the followers **406** within the portion **302***a*, the portion **302***b*, or both. Likewise, pivoting of the cover **200** may be induced by one or more actuators. Interface elements for controlling actuation of the cover **200** may be mounted to the sidewall **204**, for example, or displayed on a touch screen.

[0047] FIGS. **6**A and **6**B illustrate the cover **200** in the extended position. In the extended position, the cover **200** extends outwardly from the lip **210** (e.g., rearwardly in the X direction) and may extend outwardly from the vehicle body **102** in the plane defined by the Y and X directions. Rotation of the cover **200** in at least one direction is limited by engagement of the follower **406** with the guide **300** and support of the cover **200** by the lip **210**. For example, counterclockwise rotation (e.g., about an axis parallel to the Y direction) is prevented in the orientation shown in FIG. **6**B.

[0048] In the extended position, the cover **200** may be available to perform various functions such as:

- **[0049]** A seat for a person.
- **[0050]** A table for serving or preparing food.
- **[0051]** A platform for loading items that can then subsequently be slid into the vehicle **100** in a reverse of the process shown in FIG. **5**.

[0052] Furthermore, in the extended position, at least a portion of the storage area **214** may be accessible for removing items from the storage area **214** or loading items into the storage area **214**.

[0053] FIGS. **7**A and **7**B illustrate the cover **200** in the open position. To transition the cover **200** from the closed position to the open position, a user grasps the handle **218** and lifts upwardly, e.g., at least a portion of force exerted along the Z direction. The cover **200** may pivot (e.g., counterclockwise about an axis parallel to the Y direction) such that the follower **406** pivots within the portion **302***b* toward the inlet **304** and possibly exits the inlet **304**. In the open position, a forward edge of the cover **200** may rest on the load floor, such as area **202***c,* or on the shelf **312** or other structure on or around the load floor. When in the open position, the storage area **214** is accessible for inserting or removing items. From the open position, a user may also remove the cover **200** entirely.

[0054] Referring to FIGS. **8**A and **8**B, in some embodiments, structures may be used to retain the cover **200** in the closed or extended positions. A retention feature **800***a* may be positioned at or near a position of the follower **406** when the cover **200** is in the extended position and a retention feature **800***b* may be positioned at or near a position of the follower **406** when the cover **200** is in the closed position. For example, the follower **406** may be required to move past retention feature **800***a* to move into and out of the extended position and be required to move past the retention feature **800***b* to move into and out of the closed position. In some embodiments, the retention features **800***a,* **800***b* may impose a requirement for force to be exerted to move the follower **406** out of the extended and closed positions, respectively. In some embodiments, the retention features **800***a,* **800***b* may impose a requirement for force to be exerted to move the follower **406** more than a predetermined distance from the extended and closed positions, respectively. For example, the retention features **800***a,* **800***b* may permit 0, 1, 2, or more centimeters of movement away from the extended and closed positions, respectively, before engaging the followers **406**. The retention features **800***a,* **800***b* may require a force of at least 20, 30, 40, or more Newtons to move the follower **406** past the retention features **800***a,* **800***b*.

[0055] Referring specifically to FIG. **8**A, the retention features **800***a,* **800***b* may be implemented as protrusions **802***a,* **802***b* extending into the portions **302***a,* **302***b* that require deformation or deflection of the protrusions **802***a,* **802***b* and/or follower **406** to move past the retention features **800***a,* **800***b*.

[0056] Referring specifically to FIG. **8**B, the retention features **800***a,* **800***b* may be implemented as a spring-loaded detent **804**. For example, the follower **406** may define an indentation **806**. The detent **804** may be mounted within the guide **300** with the detent sliding in the Y direction. The detent **804** may be positioned to be urged into the indentation **806** by a spring **808** when the follower **406** is in the extended position (retention feature **800***a*) or the closed position (retention feature **800***b*). The detent **804** will slide out of the indentation **806** when the follower **406** is urged out of the extended position or closed position with sufficient force to overcome the biasing force of the spring **808**.

[0057] Referring to FIGS. **9**A, **9**B, and **9**C, in some embodiments, one or more features may be present to retain the cover **200** when in the extended position. In particular, one or more features may be present to hinder the cover **200** from moving from the extended position partially or completely toward the closed position.

[0058] Referring specifically to FIG. **9**A, the cover **200** may include a seat **900** sized, shaped, and configured to engage the lip **210**. For example, the seat **900** may include a surface **902** that faces, at least partially, along an axis parallel to the Y direction in a forward direction, e.g., in the direction that the cover **200** moves when transitioning from the extended position to the closed position. Stated differently, normal vectors along a major portion of the surface **902** have a component pointing along the Y direction in the direction moved by the cover **200** when transitioning from the extended position to the closed position (e.g., the forward direction). The normal vectors may further have a component downward along the X direction. For example, the normal vectors may define an angle of between −30 and −90 degrees relative to the X direction in the plane defined by the X and Z directions.

[0059] In the illustrated embodiment, the surface **902** is straight in the plane defined by the X and Z directions. In other embodiments, the surface **902** may be curved. The surface **902** may be curved in the plane defined by the X and Y directions to conform to curvature of the lip **210** at the bottom of the opening **212**.

[0060] Referring specifically to FIGS. **9**B and **9**C, in some embodiments, a retention feature **910** may releasably secure the cover **200** to the lip **210** either alone or in combination with the surface **902**. For example, the lip **210** may define an opening **912**, and a retention clip **914** may be secured to the cover **200**, such as a downward facing surface of the cover. The retention clip **914** may be positioned to be insertable within the opening **912** when the cover **200** is in the extended position. The retention clip **914** may be made of metal or plastic. In the absence of a deforming force, at least a portion of the retention clip **914** may be larger than the opening **912**. Accordingly, deformation of the retention clip **914** may be required to insert the retention clip **914** into the opening **912** with biasing force exerted by the retention clip **914** resisting removal from the opening **912**. In particular, the retention clip **914** may resist raising of the cover **200** from the lip **210** along the Z direction as well as sliding of the cover **200** in the X direction from the extended position toward the closed position. As shown in FIG. **9**C, there may be two openings **912** and retention clips **914** offset from one another in the Y direction, such as by at least 40, 60, or 80 percent of a width of the cover **200** in the Y direction.

[0061] The retention clips **914** and the openings **912** may cooperate with the surface **902** to resist transitioning of the cover **200** from the extended position toward the closed position: the cover **200** must be raised upwardly in the Z direction to disengage the surface **902** from the lip **210**, and the retention clips **914** resists this upward movement.

[0062] In some embodiments, the retention clips **914** may be substituted for pins or other structures that insert within the openings **912** and resist translation from the extended position toward the closed position along the X direction, but do not resist removal form the openings **912**. In some embodiments, the retention clips **914** may engage openings formed on the shelf **312** or other structure to retain the cover **200** in the closed position.

[0063] Referring to FIGS. **10**A and **10**B, a gap may be present between sides of the cover **200** and interior panels **1000** of the vehicle **100**, e.g., the wheel well covers **206** and/or sidewalls **204** in the illustrated embodiment. The gap may be present along the Y direction to permit sliding of the cover **200** relative to the interior panels **1000** in the X direction. However, such a gap permits some rotation of the cover **200** about an axis that is substantially (e.g., within 10 degrees of) parallel to the Z direction. Such rotation can result in binding of corners of the cover **200** relative to the interior panels **1000**. In some embodiments, rollers **1002** may be secured at or near front corners of the cover **200**. As used herein "front corner" may be characterized as an intersection of a front edge of the cover **200** extending substantially (e.g., within two degrees) in the Y direction with a side edge of the cover **200** extending substantially (e.g., within two degrees) in the X direction, the front edge being closer to the front of the vehicle **100** when cover **200** is in the closed position than an opposite edge of the cover **200** that also extends substantially in the Y direction ("the rear edge"). As used herein a "front corner" may include a rounded transition between the front edge and the side edge. As used herein "at or near" may be understood as (a) a distance between the roller **1002** and the front edge being 1 D or less, where D is the diameter of the roller **1002**, (b) within 5, 3, or 1 percent of a distance between the front edge and the rear edge, (c) within N*X of the front edge, where N is 3, 2, or 1, and X is a size of the gap between the side edge and the interior panel **1000** interfacing with the side edge assuming the cover **200** is centered between the interior panels **1000** on either side of the cover **200**.

[0064] As shown in FIG. **10**B, the roller **1002** may seat within a pocket **1004** formed by the cover **200** or secured thereto. The roller **1002** may be a spherical roller or a cylindrical roller with an axis of rotation substantially (e.g., within 2 degrees of) parallel to the Z direction. The roller **1002** may be held in place by a cover plate **1006** through which a portion of the roller **1002** protrudes, an axle of the roller **1002** being secured to the cover **200**, or other retainer. The roller **1002** may be biased outwardly such that pressure on the roller **1002** to retract partially into the pocket **1004**. In other embodiments, contact with the interior panel of the vehicle is not necessarily maintained such that biasing is not present.

[0065] FIGS. **10**A and **10**B show the left side of the vehicle **100** with the understanding that the right side may be in a mirrored configuration with a corresponding roller **1002** engaging the corresponding interior panel on the right side of the cover **200**.

[0066] The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

[0067] In the preceding, reference is made to embodiments presented in this disclosure. However, the scope of the present disclosure may exceed the specific described embodiments. Instead, any combination of the features and elements, whether related to different embodiments, is contemplated to implement and practice contemplated embodiments. Furthermore, although embodiments disclosed herein may achieve advantages over other possible solutions or over the prior art, the embodiments may achieve some advantages or no particular advantage. Thus, the aspects, features, embodiments and advantages discussed herein are merely illustrative.

[0068] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

What is claimed is:

1. A vehicle comprising:

a vehicle body defining a sub-floor storage area;

a cover configured to cover the sub-floor storage area and defining at least part of a load floor over the sub-floor storage area; and

guides secured to the vehicle body and defining paths configured to guide movement of the cover between:

a closed position in which the cover extends over the sub-floor storage area;

an extended position in which the cover remains engaged with the guides and extends outward from the vehicle body; and

an open position in which the cover is pivoted upwardly relative to the closed position.

2. The vehicle of claim **1**, wherein the guides include grooves, the cover having followers secured thereto and configured to engage the grooves.

3. The vehicle of claim **2**, wherein the followers include at least one of rollers or sliders.

4. The vehicle of claim **2**, wherein the guides include first portions configured to guide the followers to the extended position and second portions configured to receive the followers when in the closed position.

5. The vehicle of claim **4**, wherein the guides define inlets permitting insertion of the followers into the guides.

6. The vehicle of claim **5**, wherein the inlets are at junctions of the first portions and the second portions.

7. The vehicle of claim **1**, wherein:

the guides are molded into sidewalls of the sub-floor storage area; and

rollers are mounted to right and left sides of the cover at a forward edge of the cover and configured to engage the sidewalls responsive to twisting of the cover relative to the sidewalls.

8. The vehicle of claim **1**, wherein the vehicle body includes one or more areas defining the load floor with the cover, the cover positioned substantially flush with the one or more areas when in the closed position.

9. The vehicle of claim **1**, wherein sidewalls of the sub-floor storage area define shelves for supporting the cover when in the closed position.

10. The vehicle of claim **1**, further comprising retention features configured to resist movement of the cover out of at least one of the closed position or the extended position.

11. The vehicle of claim **1**, wherein the vehicle body defines a lip along the load floor, the cover configured to rest on the lip when in the extended position.

12. The vehicle of claim **11**, wherein the cover defines a seat configured to engage the lip and resist movement from the extended position toward the closed position when the cover is in the extended position.

13. The vehicle of claim **11**, wherein the cover has one or more retention features secured thereto and configured to releasably secure the cover to the lip.

14. The vehicle of claim **1**, wherein the cover defines a recess and a handle positioned in the recess.

15. A vehicle component comprising:

a cover configured to cover a sub-floor storage area defined by a vehicle body;

a first follower extending from a first side of the cover and configured to engage a first guide secured to a sidewall of the sub-floor storage area; and

a second follower extending from a first side of the cover and configured to engage a second guide secured to a sidewall of the sub-floor storage area.

16. The vehicle component of claim **15**, wherein the first and second followers each include a roller.

17. The vehicle component of claim **15**, wherein the first follower is secured to the cover by a first bracket, and the second follower is secured to the cover by a second bracket.

18. The vehicle component of claim **15**, wherein the cover defines a recess, and a handle is positioned within the recess.

19. A method comprising:

moving a cover from a closed position to an extended position, the cover being positioned over a sub-floor storage area defined by a vehicle body when in the closed position and extending outwardly from the vehicle body when in the extended position;

wherein moving the cover from the closed position to the extended position comprises pivoting the cover from the closed position and sliding the cover to the extended position.

20. The method of claim **19**, further comprising:

starting with the cover in the closed position, pivoting the cover upwardly into an open position in which the sub-floor storage area is accessible.

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